



CommilK

Buffalo Dairy Management System

Complete Product & Technical Specification

Web App + Mobile App | React Vite · FastAPI · Supabase · LLM AI

Project CommilK	Version 1.0 MVP	Target Dairy Farmers	Stack React + FastAPI
---------------------------	---------------------------	--------------------------------	---------------------------------

1. Project Overview

CommilK ek comprehensive dairy management platform hai jo specifically buffalo farmers ke liye banaya gaya hai. Is system mein 3 alag panels honge — User (farmer), Distributor, aur Malik (owner/admin). Saath mein AI-powered assistant aur recommendation engine bhi hoga jo yield improve karne mein help karega.

Core Problem Statement

Buffalo dairy farmers ke paas koi digital system nahi hota — milk logs, sales, health records, aur payments sab manual ya memory se track hote hain. CommilK yeh sab ek platform par laata hai.

1.1 Project Goals

- Har buffalo ka milk yield track karna — morning aur evening session alag
- Sales, payments aur buyers ka hisaab rakhna
- Buffalo health records, vaccination, aur breeding cycle manage karna
- AI assistant se farming tips aur yield improvement advice lena
- Distributor ko delivery aur collection manage karne dena
- Malik ko poora business ka overview ek jagah dekhna

2. Technology Stack

Layer	Technology	Purpose
Frontend Web	React + Vite + TypeScript	Web application

Mobile	PWA (Phase 1) → React Native (Phase 2)	Mobile users
UI Library	Tailwind CSS + shadcn/ui	Component styling
Backend	FastAPI (Python 3.11+)	REST API server
Database	Supabase (PostgreSQL)	Primary database
Auth	Supabase Auth (JWT)	Authentication
Storage	Supabase Storage	Images, documents
Primary LLM	Groq API (Llama 3.3 70B)	AI chat assistant
Fallback LLM 1	70B open-source (Ollama/Together AI)	If Groq fails
Fallback LLM 2	8B model	Secondary fallback
Fallback LLM 3	7B model (Mistral)	Last fallback
ML / Recommendations	scikit-learn + pandas	Yield & feed tips
Deployment — Frontend	Vercel	Web hosting
Deployment — Backend	Render / Railway	API hosting
Caching	Redis (Phase 2)	Performance

3. Three User Panels

CommilK mein teen bilkul alag panels honge. Har panel ka apna login, apna dashboard, aur apne permissions honge. Backend pe Role-Based Access Control (RBAC) enforce hoga.

Panel 1 — User (Buffalo Farmer)

Daily operations | Milk logging | Own buffalo management

3.1 User Panel Features

Dashboard	Aaj ka total milk yield, last 7 days graph, income summary
Buffalo Management	Apni buffaloes ki list, ek-ek ka profile, status (active/dry/pregnant)
Milk Logging	Morning + evening yield entry per buffalo, historical view
Sales Record	Kitna doodh becha, kisko, kitne mein — buyer ke hisaab se
Health Records	Vaccination dates, illness log, vet visits

Breeding Tracker	Heat cycle, insemination dates, pregnancy status
AI Chat	Apni buffaloes ke baare mein AI se sawaal karo, tips lo
Notifications	Vaccination due, low yield alert, payment reminder
Reports	Monthly income report, per-buffalo yield comparison, PDF export

3.1.1 User Access Restrictions

- Sirf apni buffaloes dekh sakta hai — dusre users ki data nahi
- Distributor panel access nahi hoga
- Malik panel bilkul access nahi hoga
- Other users ka data kabhi nahi dikhega — RLS (Row Level Security) se enforce hoga

Panel 2 — Distributor

Collection management | Multiple farmer coordination | Delivery tracking

3.2 Distributor Panel Features

Farmer Overview	Assigned farmers ki list, unka daily collection summary
Collection Log	Har farmer se kab kitna doodh liya, entry aur history
Payment Management	Farmers ko payment kab ki, kitni pending hai, history
Route Management	Delivery route plan, area-wise collection schedule
Quality Records	Fat %, SNF %, quality grade per collection
Inventory	Total collected milk, dispatched to buyer, balance
Reports	Daily/weekly collection report, payment summary, farmer-wise
Notifications	Pending payment alerts, farmer new entry approvals

3.2.1 Distributor Access Restrictions

- Sirf assigned farmers ka data dekh sakta hai
- User panel ki personal health records nahi dekh sakta
- Malik ke financial reports nahi dekh sakta
- Other distributors ki territory nahi dekh sakta
- Users ko add/remove karne ka adhikar nahi — yeh Malik ka kaam hai

Panel 3 — Malik (Owner / Super Admin)

Complete business control | All data access | Analytics & settings

3.3 Malik Panel Features

Master Dashboard	Poore business ka overview — total farms, daily production, revenue
User Management	Users add/remove/suspend, role assign karo
Distributor Management	Distributors add karo, territory assign karo, performance dekho
Business Analytics	Revenue trends, top/bottom performing farmers, seasonal analysis
Financial Reports	Total income, expenses, net profit, tax-ready summaries
Pricing Control	Milk rate set karo, seasonal price changes, buyer contracts
AI Insights	Business-level AI analysis — yield trends, market predictions
System Settings	App configuration, notification rules, backup settings
Audit Logs	Kaun kab kya kiya — complete activity trail
Export & Reports	Excel/PDF export, custom date range, all-user data

3.3.1 Malik Access

- Poora system access — koi restriction nahi
- Sabke data dekh sakta hai — users, distributors, buffaloes
- Only Malik users ko role assign/revoke kar sakta hai
- System-wide settings sirf Malik change kar sakta hai
- Audit logs sirf Malik dekh sakta hai

4. Database Schema (Supabase / PostgreSQL)

Supabase mein PostgreSQL database use hoga. Row Level Security (RLS) policies har table pe lagegi. Neeche har table ka structure diya gaya hai:

Core Tables

profiles

id	uuid (FK → auth.users, primary key)
role	enum: 'user' 'distributor' 'malik'
full_name	text NOT NULL
phone	text
village	text
district	text
avatar_url	text (Supabase Storage path)

is_active	boolean DEFAULT true
created_at	timestampz DEFAULT now()

buffaloes

id	uuid PRIMARY KEY DEFAULT gen_random_uuid()
owner_id	uuid FK → profiles.id
name	text NOT NULL
tag_number	text UNIQUE (ear tag / ID number)
breed	text (Murrah, Nili-Ravi, Surti, etc.)
date_of_birth	date
status	enum: 'milking' 'dry' 'pregnant' 'sold' 'dead'
purchase_date	date
purchase_price	numeric
image_url	text
notes	text
created_at	timestampz DEFAULT now()

milk_logs

id	uuid PRIMARY KEY DEFAULT gen_random_uuid()
buffalo_id	uuid FK → buffaloes.id
logged_by	uuid FK → profiles.id
log_date	date NOT NULL
morning_qty_liters	numeric(6,2) DEFAULT 0
evening_qty_liters	numeric(6,2) DEFAULT 0
total_qty_liters	numeric GENERATED ALWAYS AS (morning + evening) STORED
fat_percent	numeric(4,2)
snf_percent	numeric(4,2)
notes	text
created_at	timestampz DEFAULT now()

sales

id	uuid PRIMARY KEY DEFAULT gen_random_uuid()
-----------	--

seller_id	uuid FK → profiles.id
buyer_name	text NOT NULL
sale_date	date NOT NULL
quantity_liters	numeric(8,2) NOT NULL
price_per_liter	numeric(6,2) NOT NULL
total_amount	numeric GENERATED ALWAYS AS (qty * price) STORED
payment_status	enum: 'pending' 'partial' 'paid'
paid_amount	numeric DEFAULT 0
distributor_id	uuid FK → profiles.id (nullable)
notes	text

health_records

id	uuid PRIMARY KEY DEFAULT gen_random_uuid()
buffalo_id	uuid FK → buffaloes.id
record_date	date NOT NULL
record_type	enum: 'vaccination' 'illness' 'checkup' 'deworming' 'other'
description	text NOT NULL
vet_name	text
cost	numeric
next_due_date	date

breeding_records

id	uuid PRIMARY KEY DEFAULT gen_random_uuid()
buffalo_id	uuid FK → buffaloes.id
heat_date	date
insemination_date	date
method	enum: 'natural' 'AI' (artificial insemination)
pregnancy_confirmed	boolean DEFAULT false
expected_calving_date	date
actual_calving_date	date
calf_gender	enum: 'male' 'female' 'unknown'

distributor_collections

id	uuid PRIMARY KEY DEFAULT gen_random_uuid()
distributor_id	uuid FK → profiles.id
farmer_id	uuid FK → profiles.id
collection_date	date NOT NULL
quantity_liters	numeric(8,2) NOT NULL
fat_percent	numeric(4,2)
rate_per_liter	numeric(6,2)
amount_due	numeric
payment_status	enum: 'pending' 'paid'

ai_chat_logs

id	uuid PRIMARY KEY DEFAULT gen_random_uuid()
user_id	uuid FK → profiles.id
session_id	uuid
role	enum: 'user' 'assistant'
message	text NOT NULL
model_used	text (which LLM responded)
created_at	timestampz DEFAULT now()

5. Backend API Structure (FastAPI)

Saari APIs /api/v1/ prefix se start hongi. JWT token har protected route pe verify hoga. Role check middleware available hoga.

Endpoint	Method + Path	Access
Login	POST /api/v1/auth/login	Public
Register	POST /api/v1/auth/register	Public
Logout	POST /api/v1/auth/logout	Authenticated
Get Profile	GET /api/v1/profile/me	Authenticated
List Buffaloes	GET /api/v1/buffaloes	User+
Add Buffalo	POST /api/v1/buffaloes	User+
Update Buffalo	PUT /api/v1/buffaloes/{id}	User+

Log Milk	POST /api/v1/milk-logs	User+
Get Milk Logs	GET /api/v1/milk-logs?date=&buffalo_id=	User+
Add Sale	POST /api/v1/sales	User+
Get Sales	GET /api/v1/sales	User+
Health Record	POST /api/v1/health-records	User+
Breeding Record	POST /api/v1/breeding-records	User+
Add Collection	POST /api/v1/distributor/collections	Distributor+
Farmer List	GET /api/v1/distributor/farmers	Distributor+
Payment Records	GET/POST /api/v1/distributor/payments	Distributor+
All Users	GET /api/v1/admin/users	Malik only
User Roles	PUT /api/v1/admin/users/{id}/role	Malik only
Business Stats	GET /api/v1/admin/analytics	Malik only
Audit Logs	GET /api/v1/admin/audit-logs	Malik only
AI Chat	POST /api/v1/ai/chat	Authenticated
Recommendations	GET /api/v1/ai/recommendations/{buffalo_id}	User+
Daily Report	GET /api/v1/reports/daily	User+
Monthly Report	GET /api/v1/reports/monthly	User+

6. AI System Architecture

6.1 LLM Fallback Chain

Reliability ensure karne ke liye cascading fallback chain implement ki jaayegi:

Priority	Model	Provider	Condition
1 — Primary	Groq Llama 3.3 70B	Groq API	Default — fastest
2 — Fallback	Llama 3 70B	Together AI / Ollama	Groq fails/timeout
3 — Fallback	Llama 3.1 8B	Together AI	70B unavailable
4 — Fallback	Mistral 7B	Together AI	8B also fails

6.2 AI System Prompt Context

Har AI request mein user ka context inject hoga — unki buffaloes ki info, recent milk logs, aur location. Isse AI personalized aur relevant answers dega.

6.3 Recommendation Engine

- Feed Recommendation — milk yield history dekhke optimal feed quantity suggest karna
- Health Alert — agar buffalo ka yield achanak 20%+ gire toh health check alert
- Breeding Suggestion — heat cycle pattern se optimal breeding time predict karna
- Market Tip — local milk rates ke hisaab se best selling time suggest karna
- Seasonal Advice — garmi/sardi ke hisaab se management tips

Phase 1 mein rule-based logic se start karo. Phase 2 mein actual scikit-learn model train karo historical data pe.

7. Security Architecture

7.1 Authentication

Auth Provider	Supabase Auth (email + phone OTP)
Token Type	JWT with 24hr expiry + refresh token
Session	Supabase managed sessions
Role Storage	profiles.role column — checked on every request

7.2 Row Level Security (RLS) Rules

- buffaloes — User sirf owner_id = auth.uid() wali rows dekh/edit kar sakta hai
- milk_logs — User sirf apni buffaloes ke logs access kar sakta hai (via buffalo join)
- sales — Seller sirf apni sales, Distributor assigned sales, Malik sab
- health_records — Sirf buffalo owner ya Malik access kar sakta hai
- profiles — Har koi apna profile dekh sakta hai, Malik sab dekh sakta hai
- distributor_collections — Distributor apni collections, Malik sab

7.3 Rate Limiting

Login API	5 requests per minute per IP
AI Chat API	20 requests per minute per user
General APIs	100 requests per minute per user
File Upload	10 requests per minute per user

7.4 Input Validation & Security

- Pydantic models se saari inputs validate hongi — FastAPI automatically
- Supabase parameterized queries use hongi — SQL injection impossible

- File uploads mein MIME type check + size limit (images: 5MB, docs: 10MB)
- CORS sirf allowed origins se — frontend domain whitelist
- Environment variables mein secrets — kabhi code mein nahi
- HTTPS only production mein — Vercel aur Render automatically handle karte hain

8. Supabase Storage Buckets

buffalo-images	Public bucket — buffalo photos (public read, owner write only)
profile-images	Public bucket — user profile pictures
health-docs	Private bucket — vet prescriptions, documents (owner + malik only)
reports	Private bucket — generated PDF reports (user-specific access)

Storage policies Supabase RLS se link hongi. Private bucket files sirf signed URLs se access hongi — direct public URL nahi milega.

9. Project Folder Structure

Frontend: /frontend

```
frontend/ ├── src/ |   ├── pages/ |   |   ├── user/           ← User panel pages |   |   ├── distributor/ ← Distributor panel pages |   |   ├── malik/         ← Malik panel |   |   ├── shared/      ← Common UI components |   |   ├── components/ |   |   ├── charts/        ← Dashboard graphs |   |   ├── forms/        ← Input forms |   |   ├── hooks/       ← Custom React hooks |   |   ├── services/     ← API call functions |   |   ├── stores/      ← Zustand state management |   |   ├── utils/        ← Helper functions |   |   ├── types/       ← TypeScript interfaces |   ├── public/ |   ├── .env.local |   └── vite.config.ts
```

Backend: /backend

```
backend/ ├── app/ |   ├── main.py           ← FastAPI app entry point |   ├── routes/          |   ├── auth.py          |   ├── buffaloes.py     |   ├── milk_logs.py     |   ├── sales.py         |   ├── health.py        |   ├── breeding.py      |   ├── distributor.py   |   ├── admin.py         |   ├── ai_chat.py       |   ├── reports.py       |   ├── models/          ← Pydantic |   ├── request/response schemas |   ├── services/       |   ├── llm_service.py   ← Groq + |   ├── fallback chain |   ├── rec_engine.py   ← Recommendation logic |   ├── report_gen.py    ← PDF/Excel reports |   ├── middleware/     |   ├── auth_middleware.py |   ├── rate_limiter.py |   ├── db/             |   ├── supabase_client.py |   ├── utils/          |   ├── validators.py    |   ├── helpers.py      |   ├── .env             |   └── requirements.txt
```

10. Development Phases

Phase 1 — MVP (4 Buffalo, Abhi Shuru Karo)

Target: 4-6 weeks | Basic working system

- Supabase project setup + all tables + RLS policies
- FastAPI backend — auth, buffaloes, milk_logs, sales APIs
- Frontend — Login, Dashboard, Buffalo list, Milk log entry
- User panel complete — basic features only
- Malik panel — user list, basic analytics
- AI Chat — Groq integration, simple context
- Deployment — Vercel + Render live

Phase 2 — Full System

Target: 8-12 weeks | All 3 panels complete

- Distributor panel complete
- Recommendation engine (scikit-learn)
- Health alerts + breeding tracker
- PDF/Excel report generation
- Advanced analytics for Malik
- PWA mobile support
- Payment tracking complete

Phase 3 — Scale Karo

Future: When business grows

- React Native mobile app (Android + iOS)
- Multiple farm support / franchise model
- IoT milk meter integration
- WhatsApp bot for quick milk entry
- Government scheme integration
- Redis caching layer
- Multi-language support (Hindi, Punjabi, Gujarati)

11. Environment Variables

Backend .env

```
# Supabase SUPABASE_URL=https://your-project.supabase.co  
SUPABASE_SERVICE_ROLE_KEY=your_service_role_key SUPABASE_ANON_KEY=your_anon_key # LLM —
```

```
Primary GROQ_API_KEY=your_groq_api_key # LLM - Fallbacks
TOGETHER_API_KEY=your_together_ai_key # App SECRET_KEY=your_jwt_secret_minimum_32_chars
ENVIRONMENT=development ALLOWED_ORIGINS=http://localhost:5173,https://yourdomain.com
API_VERSION=v1
```

Frontend .env.local

```
VITE_API_BASE_URL=http://localhost:8000/api/v1 VITE_SUPABASE_URL=https://your-
project.supabase.co VITE_SUPABASE_ANON_KEY=your_anon_key
```

12. Instructions for AI Code Generation

Yeh document AI (Claude / GPT) ko dena hai

Jab bhi code generate karao, pehle yeh document share karo

Jab tum kisi AI se code generate karo, tab yeh bolo:

```
"Read this CommilK specification document carefully. Build the [FEATURE NAME] for
this system following: - Tech stack exactly as specified (React Vite + FastAPI +
Supabase) - 3 panels: User, Distributor, Malik with RBAC - All security rules
including RLS and JWT auth - Database schema exactly as given - API routes as
specified - Production-ready, no partial code"
```

12.1 Feature Build Order (Recommended)

1. Supabase schema + RLS policies SQL file
2. FastAPI backend — main.py + auth routes + Supabase client
3. Buffalo CRUD + Milk log APIs
4. Frontend — Auth pages (login/register)
5. Frontend — User Dashboard + Buffalo Management
6. Frontend — Milk Log entry + Sales
7. AI Chat (Groq + fallback) — backend + frontend
8. Distributor Panel — collections + farmer list
9. Malik Panel — user management + analytics
10. Recommendation engine + health alerts
11. Reports (PDF + Excel export)

CommilK — Apne 4 Bhainso se shuru karo, baad mein hazaro tak scale karo.

Yeh document poori project ki neev hai. Isko root folder mein CommilK_Specification.docx ke naam se rakh do.

